DTN Routing: Secure Message Transmission in Border Defence System

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***Abstract*— Delay tolerant networks have already evolved as a pragmatic solution to various requirements such as delayed routing in wireless networks. They can also be used in place of traditional wireless networks in the border areas or highly sensitive areas, thus contributing to the data security in such places. In this paper, a border security architecture has been proposed where DTN is used for transferring messages within the army units of a particular region that are deployed in harsh terrains, thereby ensuring integrity and security of the message. In the proposed approach, a special property of the nodes is used which makes them self disposable in nature after either their mentioned TTL gets over or whenever data tampering takes place in the node. So, the message as well as the node will be definitely safe from getting accessed in an unauthorized manner.**

Keywords— ***DTN, routing, security, bio degradable node, architecture***

I. INTRODUCTION

Delay tolerant networks are an indispensible part of the wireless networks community that is characterized by frequent disconnections when the transfer of message takes place. This is one major property of the DTNs that isolate them from other wireless type connections. The disruption in the delay tolerant network varies depending upon the geographical terrains, size of message, transmission rate of the node et al. In a DTN, each node has the peculiar ability to store data packets and then forward them using opportunistic contact. But security of the message is vulnerable in such networks owing to a large number of intermediate hops before reaching the destination. In the below presented work, a border security scenario has been shown where the communication among army units is protected from any unauthorized access. For the security of message, an encryption technique is used which encrypts the message inside a node before it is sent to the destination region. The technique has been explained in the given text. But in such cases, data security as well as security of the DTN node, both are vital. To address such a security issue, the use of DTN nodes made up of biodegradable material has been proposed. For the sake of simplicity and convenience, rest of the paper is organized as follows: Section 2 below shows the problem scenario prevailing in the border areas followed by section 3 which presents the motivation of doing this work. Section 4, 5 and 6 show the related work and proposed approach to the problem, followed by the conclusion and future work.

II. PROBLEM SCENARIO

These days border intrusion is a growing threat in the world. Millions of dollars are spent every year by the nations to safeguard their border regions from unidentified access and unknown encroachments. Nations are trying to intrude into their own neighbor's land in order to get hold of sensitive information or resources that may be used later on in war or in agitation. Such wars have made the army units of respective countries to be deployed in the border region to protect their territories from getting interpolated. But there have been numerous instances reported by investigating agencies of the countries regarding information being leaked or the dialogues getting intercepted when message is transmitted within the deployed army units. This might have resulted as an effect of the harsh terrains and geographical diversity in such regions. So in order to plug in these loop holes, a message transfer architecture, for securing both data and node to avoid unnecessary interjection and data pilferage, has been proposed.

Although there are various security architectures available that have been applied on the border region, but very little of absolutely no effort has been taken up to protect the node that carries message inside it. Fortifying the node is crucial because if any person gets hold it, he can easily extract the message within it and this may lead to data breach resulting in compromising the security of that nation.

III. MOTIVATION OF THE PRESENT WORK

As there have been multiple issues and concerns encountered in border security, there is a need to implement some robust security architecture that does not only protect the message but also make the message carrying node defiant to attacks. This has led to motivation of proposing a property in the DTN node that makes them self disposable in nature. As stated in [1], the minimum hop count is directly proportional to the vulnerability of node, so if the hop count increases, it makes the message more prone to attacks. In the below presented approach, node that carries message is directly dropped at the particular location by the army helicopter/airplane thereby reducing the hop count to just one. For doing this the message is first encrypted and then transferred to the designated helicopter (as explained in the algorithms in section 8).

ISBN: 978-1-5090-6471-7/17/$31.00 ©2017 IEEE

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The message nodes used are made up of a completely biodegradable material with non toxic battery as in [2] so that it doesn't cause any harm to the environment in due course of time. Such wireless sensor nodes can be procured in bulk so that the overall expenditure on such nodes is reduced and consequently reducing the budget of the security in harsh terrain areas.

IV. RELATED WORK

There has been much research work done on the security in DTNs, various attacks [17] and security considerations [9], [13] have been studied. A hierarchical identity based encryption technique is proposed in [11] which deals with fine grained revocation and access method of the messages within the DTN infra structure. A technique for message protection has been proposed in [12] called i-Trust which finds out any abnormality in the node's behavior that may take place while data transfer. Also in [10], UN peacekeeping scenario has been studied to analyze the security requirements in similar areas. Considerable research work on routing [14], [15] has also been carried out, but there have not been any significant attempts to safeguard the data and node both during message transmission.

V. SECURITY THREATS IN DTN

Major security threat in DTN [17] is due to other networking elements that act as intermediate nodes for the transfer of messages. But this problem has been overcome in the below presented approach because all the nodes are just one hop away from each other. Other attacks include masquerading, where the data packet passes through some undesired node and the data is altered by such node, thereby passing on the tampered data to the destination node. Similarly, one more attack that frequently takes place at sensitive regions such as border areas is eves dropping, where the message passes through an unwanted entity who silently senses the message and may misuse it in future.

VI. CRYPTOGRAPHIC TECHNIQUES

There are a number of cryptographic techniques [18] used today. Some of them are as follows:

1. Caesar Cipher: It is like a shift cipher where the plain text characters are shifted by three places to get the ciphertext.
2. Playfair cipher: It is like a polygraphic cipher which encrypts two characters at a time. It is more robust than the Caesar cipher.
3. Vignere Cipher: It is a poly-alphabetic substitution cipher where a single plain text letter can be converted to multiple cipher text letters.
4. Rail Fence Cipher: It is like a transposition cipher where the characters are written in a zig zag manner to obtain the ciphertext.
5. Modern Cipher: These are those techniques which use both substitution as well as transposition cipher to convert plain text to ciphertext. Examples of such techniques include DES, AES, RSA et al.

VII. PROPOSED APPROACH

The Border defence system describes a scenario where message transfer takes place through a Delay Tolerant networks established between the army units deployed on the border region of a country, say country 'A'. There are many messages or commands that have to be transferred from the commanding officer to the jawans who are deployed in the remote areas in a confidential manner. This message may be related to some operation or any other basic military exercise to be performed.

So instead of deploying conventional wireless networks, DTN can be used to save money and ensure that the messages are always secure and free from tampering. Figure 1 shows the architecture of such a system [19]. It depicts two countries that are in war mode (although it is not necessary that two neighboring countries are always in war mode, but here it is considered to give a clear perception). Consider a case where the commanding officer (CO) wishes to send a secret message or some command to his battalions (battalions here mean the unit of jurisdiction of the Commanding officer). So as per the encryption scheme, he prepares a message by encrypting it and encoding it inside a DTN node. The encryption method used in the below presented architecture is robust enough to safeguard the node from any kind of undesirable access to the node and the message. The helicopters assigned for these tasks then carry the nodes and drop them near their respective battalions. This location related information is given to the helicopter in form of latitude and longitude value of the battalion. Similarly such activities are taken up by all other COs that have their units deployed in the sensitive areas. In the meanwhile, it is considered that sensing some activity over the border, enemy team may also try to get hold of the node and retrieve the message. But as already discussed, the nodes are of sensitive nature, any type of tampering or entering wrong credentials for 2-3 times will lead to its self decomposition and thereby making the message safe from unauthorized access.

Our focus area is to deploy DTN in the border areas or war zones such that CIA of the message is maintained. Here CIA refers to the three pillars of data security: C stands for confidentiality, I stands for integrity and A stands for authenticity. To accomplish this, an encryption scheme to encrypt the message has been used along with a special type of DTN node which is self destructive or bio degradable [8] in nature. This would ensure that the DTN nodes dropped all over the border region don't eventually cause environment pollution of any sort in due course of time after their message is read by the authorized personnel.

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A. STRUCTURE OF THE NODE

The structure of the special type of DTN node described in this approach is somewhat similar to the imote2 [2] developed by Crossbow Technology Inc. The node has a net weight of around 10-12g and is a tiny networking element that has the capability of storing short messages up to 256 kb in size. This size of message is different for different nodes depending upon the size of primary memory used by them. These nodes are powered with a 256kb RAM (as per specification), a transceiver which is capable of receiving and transmitting message for not more than 4 times in its lifetime and a processor of a particular make. The power consumption of these nodes is 31mA when the radio is in off state and 44mA when it is on. The data rate is 250kb/s.

Latitude(512B)	Longitude(512B)	Drop_Time(512B)
No. Of nodes(200B)		Load_Time(512B)
Node ID(512B)		Size(300B)
TTLn(262B)	Crack_Time(262B)	CO-ID(512B)
k1(512B)		k2(512B)
Encryption(1KB)		

Table 1. Message Header

B. MESSAGE TO BE ENCODED IN THE NODE

As stated above, the major aim of this approach is to deploy DTN nodes such that they convey the message and after it is done, self destroy or start degrading. Therefore, it is paramount that the message inside the node must have a header which can comprehensively guide the receiver in this direction. Therefore message header consists of all the information regarding the Battalion including its exact location that it is meant for.

C. DESCRIPTION OF FIELDS IN THE MESSAGE HEADER

1. Latitude: This gives the latitude value of the location of battalion present on the border.
2. Longitude: This gives the longitude value of location of battalion present on the border.
3. TTL: It is the TTL of the node. It is one of the most crucial fields in the header. It is crucial because it is the field that sets the lifetime of these nodes. This time is set in such a way that if nobody comes to read the message from node, it will get destroyed after this much amount of time. It is the maximum time to live of the node.
4. Number of nodes: It is the total number of nodes present in the network.
5. Time to load: Time at which the node is loaded on the helicopter/airplane.
6. Time to drop: Time at which the node is dropped at the prescribed latitude and longitude value.

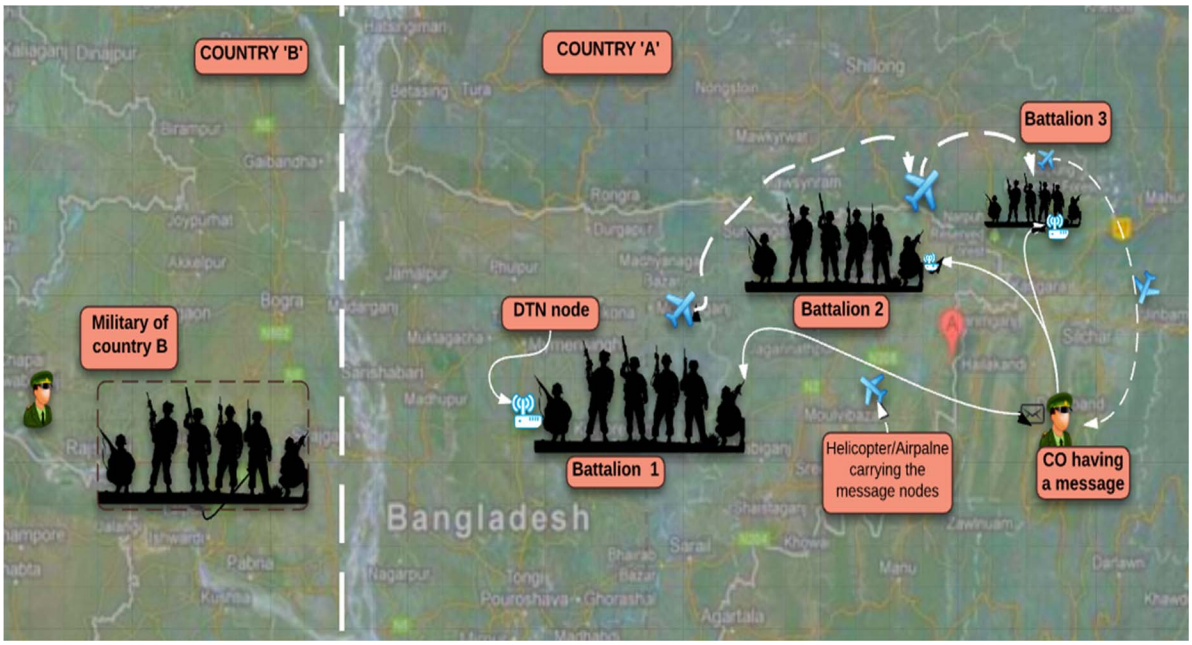

Fig. 1. Architecture of the proposed usage of Bio Degradable Nodes in Border defence system [7]

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7. Time to load: Time at which the node is loaded on the helicopter/airplane.
8. Time to drop: Time at which the node is dropped at the prescribed latitude and longitude value.
9. CO-ID: Unique ID of the commanding officer. This clearly tells about the sender of the message.
10. U-ID: Unit ID of the unit to which the CO belongs.
11. Node-ID: Unique ID of the node.
12. Size: Size of the message inside the node.
13. Key: This is the unique alphanumeric code of the node which is matched with the corresponding code of the receiver to unlock the node and retrieve the message. Two keys are used here:
    k1: This consists of only alphabets
    k2: This consists of only numbers
14. Encryption: It gives the encryption technique used. The encryption technique used to convert the plain text to cipher text is explained as follows:
    Let C be the cipher text and P be the plain text.
    C can be obtained using the equation:
    $$C = \{P + (k1+k2)\} \bmod 26$$
    For the purpose of addition, ASCII value of the key k1 is taken and added to k2.
15. Time to crack: This is the time required to crack the message of the node when key is not known.

Total size of the message header is 6kb and therefore the maximum size of the message that can be stored in the node is 250kb. So, the whole message can be retrieved in just 1 second owing to the data rate of 250kb/s. Table 1 shows the header format of the message.

VIII. ALGORITHMS TO IMPLEMENT THE BORDER DEFENCE SYSTEM

The below mentioned algorithms describe the full procedure of implementing the border defence system. For better understanding, flow chart in figure 2 can be referred.

A. Algorithm 1: Populating the node

Notation
H : header of the message
PT: plain text
Trigger: Storage of data into the node

1. Input message header fields [input(H)]

Let
l^t: Latitude value of the Battalion
l^n: Longitude value of the Battalion
t: TTL of the node
n: Number of nodes in the network
n_i: i^{th} node in the network
t_c: Time to crack message in the node, where $t_c > t$
l_t: Load time of the node
d_t: Drop time of the node
c: CO-ID
u: Unit-ID
N: node ID
m: size of the message to be sent
$k1$: Alphabetical key used to retrieve the message
$k2$: Numerical key used to retrieve the message
e: encryption technique used

2. Store header fields in the node
   $H \leftarrow \{ l^t, l^n, t, n, t_c, l_t, d_t, c, u, N, m, k1, k2, e\}$
3. Enter the message in the node [enter(PT)]
   $PT \leftarrow$ {message to be sent}

B. Algorithm 2: Obtaining the Ciphertext

Notation
CT: Ciphertext
Trigger: Encrypting the plain text message

1. Convert PT into ciphertext [encrypt($PT, k1, k2$)]
   $CT \leftarrow$ encrypt ($PT, k1, k2$)
   $x \leftarrow$ encrypt($PT, k1, k2$)
   return [{PT+ (k1+k2)} mod 26]

C. Algorithm 3: Concatenation

Notation
X: Prepared message ready to be sent to the node
Trigger: Preparation of the message

1. Concatenate H with CT [prepare_message (CT, H)]
2. $X \leftarrow$ {prepare_message (CT, H)}

D. Algorithm 4: Sending the message

Notation
Y: Node loaded with message ready to be transferred
hid: helicopter ID
Trigger: Transfer of the message

1. Align the message with the respective helicopter ID
2. $Y \leftarrow$ align(hid, X)
3. Transfer(Y)

E. Algorithm 5: Self destruction of the node

Trigger: After the node is dropped

```
if (n_i ←retrieved by authorized personnel)
        decrypt (CT, k)
        self_destroy (n_i)
elseif (n_i ←retreived by unauthorized personnel)
        if (t_c <t)
                t++
        else
                self_destroy (n_i)
else
        self_destroy (n_i)
```


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IX. WORKING EXAMPLE OF THE ALGORITHMS

Let *hi* be the message to be sent, first key be *A* and the second key be *a randomly picked up value between 0 to 9999*.

Therefore, PT = hi

$k1$ = A

$k2$ = 1024

To convert the message into ciphertext, the PT is first converted to decimal form. For this the ASCII value of individual characters is taken and then added.

Therefore, PT= 104+105=209

$k1$= 65

$k2$= 1024

As per the equation, $x = [\{PT + (k1+k2)\} \bmod 26]$

$$x = [\{209 + (65+1024)\} \bmod 26]$$
$$= [1298 \bmod 26]$$
$$x = 24$$

So, *24* will be transmitted as the ciphertext.

The ciphertext is then concatenated with the helicopter ID and sent to the desired location.

X. CONCLUSION

This paper is an extended version of the paper titled "BDN: Bio Degradable Node, Novel Approach For Routing In Delay Tolerant Network" [16]. This is the first, of its kind, approach for providing data security in DTN by using a highly robust architecture that uses simple encryption algorithm. This proposal is fool proof as no breach of the data security takes place because even if the message doesn't reach the authorized personnel, it will still not go into the hands of any unauthenticated/ unintended person. Also the message is secure because the destination is just one hop away. After the implementation of such a defence system on the border areas, the messages and sensitive information can be kept safe from all types of hackers and crackers.

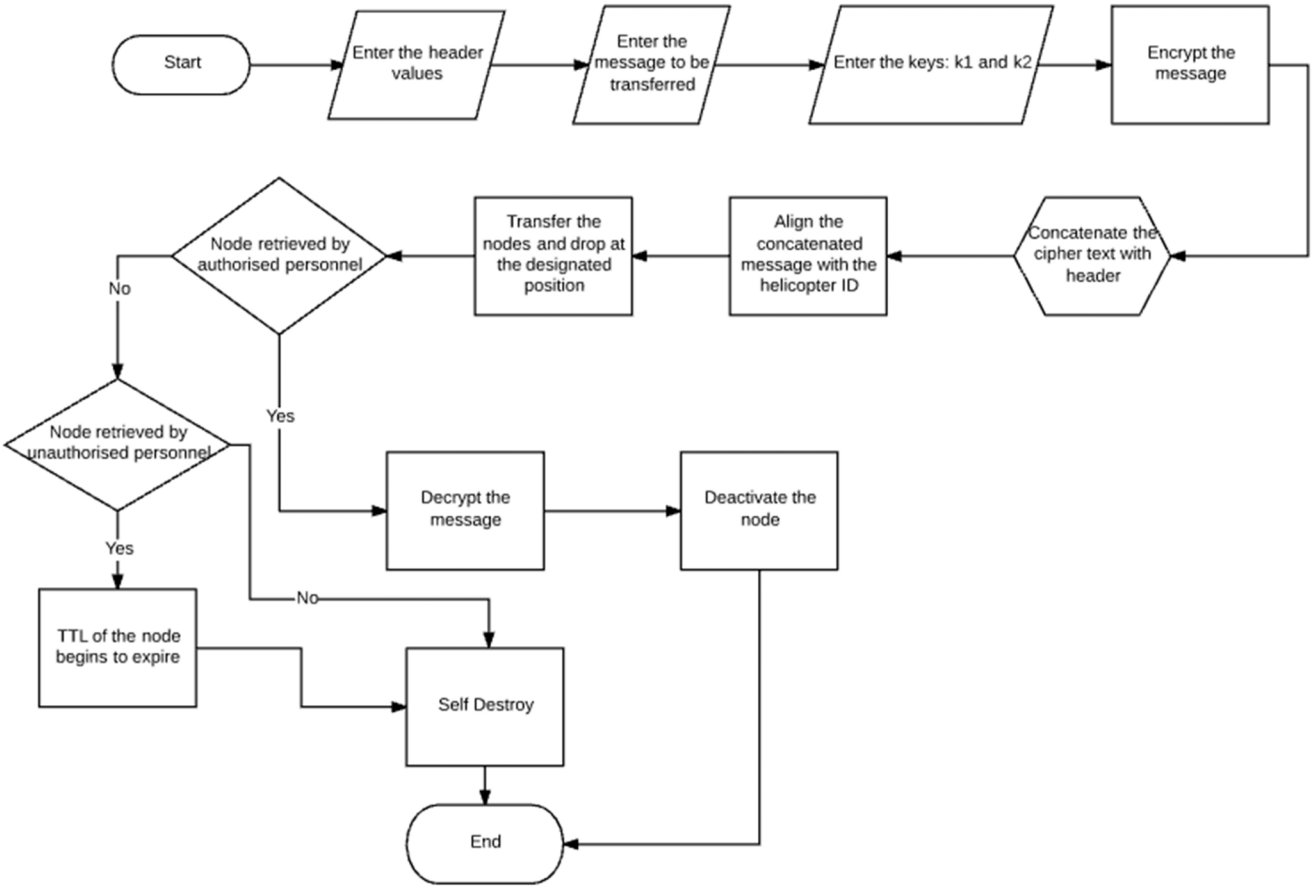

Fig. 2. Flow chart showing the implementation of the border defence system

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XI. FUTURE WORK

As part of the future work to the existing work, it is proposed to implement the algorithms described in the text using ONE [4] simulator. The messages would be encrypted using more robust algorithm at the source which would include 8 bit key encryption technique or 16 bit key encryption technique and the encrypted message would then be processed as stated above. For routing of the data packets [3], any of the routing algorithms like MaxProp [6] or Epidemic[5] would be used to transfer such messages. This architecture can be deployed by defence establishments in order to safeguard their data without disturbing their defence budget.

ACKNOWLEDGEMENT

A deep sense of gratitude is expressed towards the University School of Information and Communication Technology along with GGS Indraprastha University for providing all the necessary and relevant resources for carrying out this research work. This paper is an extended and expanded version of "BDN: Bio Degradable Node, Novel Approach For Routing In Delay Tolerant Network" presented at International Conference on Frontiers of Intelligent Computing: Theory and applications (FICTA) [16].

REFERENCES

[1]Choo, Fai Cheong, Mun Choon Chan, and Ee-Chien Chang. "Robustness of DTN against routing attacks." *Communication Systems and Networks (COMSNETS), 2010 Second International Conference on*. IEEE, 2010.

[2]http://www.xbow.jp/imote2.pdf.

[3]Afreen Fatimah, Rahul Johari, "PART: Performance analysis of routing techniques in delay tolerant networks", *International conference of internet of things and cloud computing, Cambridge, UK (ICC'16)*, ACM, 2016.

[4]Keränen, Ari, Jörg Ott, and Teemu Kärkkäinen. "The ONE simulator for DTN protocol evaluation." *Proceedings of the 2nd international conference on simulation tools and techniques. ICST*, 2009.

[5]A. Vahdat and D. Becker, "Epidemic Routing for Partially Connected Ad Hoc Networks," Technical Report CS-200006, Duke University, Durham, April 2000.

[6]El Mastapha Sammou and Abdelmounaim Abdali, "Routing in Delay Tolerant Networks(DTNs), Improved Routing with MaxProp and the Model of "Transfer by Delegation" (Custody Transfer)", Int. J. Comm, Network and System Sciences, 2011, 4, 53-58.

[7]http://www.news18.com/news/india/5-2-earthquake-jolts-india-bangladesh-border-593941.html.

[8]Anderson, Noel Wayne. "Non-toxic, biodegradable sensor nodes for use with a wireless network." U.S. Patent No. 8,063,774. 22 Nov. 2011.

[9]Farrell, Stephen, and Vinny Cahill. "Security considerations in space and delay tolerant networks." *Space Mission Challenges for Information Technology, 2006. SMC-IT 2006. Second IEEE International Conference on*. IEEE, 2006.

[10]Bhutta, N., et al. "Security analysis for delay/disruption tolerant satellite and sensor networks." (2009): 385-389.

[11]Patra, Rabin, Sonesh Surana, and Sergiu Nedevschi. "Hierarchical identity based cryptography for end-to-end security in DTNs." *Intelligent Computer Communication and Processing, 2008. ICCP 2008. 4th International Conference on*. IEEE, 2008.

[12]Punitha, K., et al. "A Secure I-Trust Scheme Towards Periodic Trust Establishment in Delay-Tolerant Networks." (2015).

[13]Ahuja, Sachin, Rahul Johari, and Chetna Khokhar. "EAST: exploitation of attacks and system threats in network." In *Information Systems Design and Intelligent Applications*, pp. 601-611. Springer India, (2015).

[14]Johari, Rahul, and Dhari A. Mahmood. "GA-LORD: Genetic Algorithm and LTPCL-Oriented Routing Protocol in Delay Tolerant Network." In *Wireless Communications, Networking and Applications*, pp. 141-154. Springer India, (2016).

[15]Johari, Rahul, and Sakshi Dhama. "Routing Protocols in Delay Tolerant Networks: Application-Oriented Survey." In *Wireless Communications, Networking and Applications*, pp. 1255-1267. Springer India, (2016).

[16] Fatimah, Afreen, and Rahul Johari. "BDN: Biodegradable Node, Novel Approach for Routing in Delay Tolerant Network." *Proceedings of the 5th International Conference on Frontiers in Intelligent Computing: Theory and Applications*. Springer, Singapore, 2017.

[17]Farrell, Stephen, and Vinny Cahill. "Security considerations in space and delay tolerant networks." *2nd IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT'06)*. IEEE, 2006.

[18]Rajput, Yashpalsingh, Dnyaneshwar Naik, and Charudatt Mane. "An Improved Cryptographic Technique to Encrypt Text using Double Encryption." *International Journal of Computer Applications* 86.6 (2014).

[19] Afreen Fatimah, Rahul Johari, "Defence Deployment System: Application of the Delay Tolerant Network", India Patent Application 201611019150, January 13, 2017.

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